

RAG:

Retrieval-Augmented Generation. Basically, when we get a file, we use the query to search through the file to find relevant information about the query, then we feed the results to an ai and the ai will give it's answer based off of the information already collected for it. RAGs try to prevent hallucination by giving only relevant information to the ai. So instead of the ai knowing everything and assuming it's right, it knows only what you want it to know about your topic.

RAGs are built on the way we determine if the query is close to items in a file. By assigning a value to each word, then seeing which other words are close to the words in the query, we find similarities. Think about the word "basketball," what words are related to "basketball?" Maybe, "soccer," "volleyball," or "sports." These words cluster together because they are found to be with a lot of the same words. Words gain meaning by defining them with other words. In this way we can expect what words are "close" to each other. If you throw in the word "flower," you may not find "basketball" and "flower" to be very related, you could even say they have a great "distance" between themselves, this concept of "distance" makes the search portion of RAGs. The greater the "distance" the more likely the words are not related to each other, the more likely they are not needed for the query.

PDF READER:

The first tab is dedicated to the Pdf reader. Upload a Pdf, wait for the chunks to be formed, then ask questions about the Pdf and you will get an answer in the container.

Upload one pdf at a time and get your answer for that pdf.

On the sidebar there is an option to change the chunks and overlap.

CHUNKS:

Chunks are chunks of text from the pdf. Changing the chunks value changes how many characters are in each chunk.

Too big of a chunk means that there is too many characters and you may get unnecessary information. Too small of a chunk means that you might miss related information.

OVERLAP:

Overlap is the overlap between chunks. We use overlap to make sure each chunk has more context.

For example, if I took a pdf and just split it by the periods, we might miss some context that the other sentences give. Overlap stops this and makes sure we gather context from other chunks too.

Too big of an overlap and each chunk will not be much different from the previous one. Too little. And there may not be enough context.

FEEDBACK SYSTEM:

It's just a system that takes in user input and saves it into a file, then prints out the previous and current feedback.

MAD-LIB MAKER:

This was just a series of text prompts and strings.